

# TONY ABRAHAM

## Embedded Systems Engineer

Phone: 8281636175 | Location: Technopark Phase 1, Thiruvananthapuram, Kerala, India |

Email: [toneyabraham37@gmail.com](mailto:toneyabraham37@gmail.com) | LinkedIn: <https://www.linkedin.com/in/toney-abraham-9bb16424a/>

### ABOUT ME

Embedded Software Engineer with 3.5 years of experience in Embedded Linux, Zephyr RTOS, low-level driver development, board bring-up, and functional safety validation. Experienced in ARM Cortex platforms, device drivers, hardware interfacing, and automation frameworks for embedded systems.

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### SKILL

- Programming Languages: C, C++, Python
- Microprocessors: ARM Cortex-A
- Microcontrollers: ARM Cortex-M, ATSAM5x/E5x, ATSAMC21, ATSAM1x, PIC18F65, ESP32
- RTOS & Frameworks: Zephyr RTOS, Device Tree, Kconfig, Board Support Package, Embedded Linux
- Protocols & Peripherals: UART, SPI, I2C, DAC
- Functional Safety: ISO 26262, IEC 60730, A-Spice, MISRA C
- Tools: GIT, GitHub, Bitbucket, MPLAB X IDE, Visual Studio Code
- Project Management: Jira, Confluence
- Testing & Debugging: Ztest, Twister (Zephyr), OpenOCD

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### EDUCATION

- Master of Science in Electronics, Mahatma Gandhi University, Kottayam (2020-2022)
- Bachelor of Science in Electronics and Computer Science, Mangalore University, Mangalore (2017-2020)

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### WORK EXPERIENCE

**Quest Global Engineering Solutions Pvt Ltd** – Embedded Software Engineer      Oct 2025 – Present

#### 1. Embedded Linux Validation

- Developed an automated CPU/Cluster power domain validation framework
- Implemented multi-thread CPU core and Cluster Power ON/OFF heat run validation using Linux CPU hot plug and monitoring the registers.
- Automated test execution, log capture, and HTML report generation using Python, reducing manual validation effort.

**Target Microprocessor:** Cortex A76

**Tech Stack:** Bash Scripting, Embedded Linux, python

**Emirates Automation Pvt Ltd** – Embedded Design Engineer

Nov 2022 – May 2025

#### 2. Zephyr Driver Development – Client: Microchip Technology Pvt Ltd

- Integrated Microchip SoCs into Zephyr RTOS through BSP, Device Tree, and driver enablement.
- Integrating Microchip device families into Zephyr's platform
- Developing and upstreaming device support for microchip peripherals in Zephyr
- Creating and maintaining Device Tree files and Kconfig entries for Microchip devices.

- Focused on board-level support package, driver abstraction and hardware validation
- Added HAL of Microchip device to Zephyr Platform

**Target Microcontrollers:** ATSAMD5x/E5x, ATSAMD1x

**Tech Stack:**

- Zephyr RTOS, Embedded C, Device Tree and Kconfig configuration
- Board Support Package and low-level driver development, Git, West build system, OpenOCD Debugging
- UART, SPI, DAC
- Visual Studio Code, MPLAB X IDE

**Additional Contribution:**

- Developed a custom Visual Studio Code extension "[zephyr-build](#)" to simplify Zephyr command execution during development
- Zephyr Upstream Contribution: [Microchip SAMD5x/E5x](#)

## 2. Aroma - Home Automation Product (In-House)

- Developed application-level firmware for a custom embedded project targeting Microchip PIC18F and Espressif ESP32 platforms.
- Validated and verified the working of the device in Real-Time with the setup.
- Managed requirement analysis, prioritized development features, and facilitated effective communication between stakeholders.
- Translated customer requirements into actionable software development tasks, ensuring alignment with product goals and timelines.

**Target Microcontrollers:** PIC18F65, PIC18F66, ESP32-DevKit.

**Tech Stack:**

- C Programming, MPLAB X IDE, ESP-IDF
- I2C, USART, RTC, ADC Integration
- Debugging with UART, Logic Analyzer, Git, Embedded Systems Debugging

## 3. ClassB Library development and validation – Client: Microchip Technology Pvt Ltd

- Developed and validated a Class B Software Library to comply with IEC 60730 safety standards
- Implemented critical startup and runtime tests: CPU register test, Program counter test, Variable memory test, EEPROM/Flash memory test, Clock test
- Ensured functional safety using MISRA C guidelines and hardware fault injection methods
- Used automated and manual unit testing and code coverage tools for validation

**Target Microcontroller:** PIC32CMJH

**Skills and Tools:**

- C Programming, MPLAB X IDE, Git, GitHub, Jira, Confluence, Functional Safety Standards (IEC 60730), MISRA C

**Additional Contribution:**

- ClassB Contribution: [CLASSB\\_PIC32CMJH\\_Library](#)

## 4. Functional Safety Validation – Client: Microchip Technology Pvt Ltd

- Verification and Validation of ATSAMC21 and ATSAMV71 as per ISO 26262
- Followed **ASPICE V-Model** development process for plan, implementation and validation
- Developed **test scenarios** and **test cases** aligned with **ASIL-B** functional safety requirements
- Conducted **Qualification Test**, **Integration Test** and **Unit Test** across the Software requirement, Architectural Design and Detailed Design.
- Contributed to the safety analysis documents: SWE.4, SWE.5, SWE.6

**Target Microcontroller:** ATSAMC21 and ATSAMV71

**Tech Stack:**

- C Programming, Git, Bitbucket, Functional Safety Checklist, Static Code Analysis
- Vector Cast, MPLAB X IDE, Jira, Confluence, Polarion
- ISO 26262, ASPICE V-Model